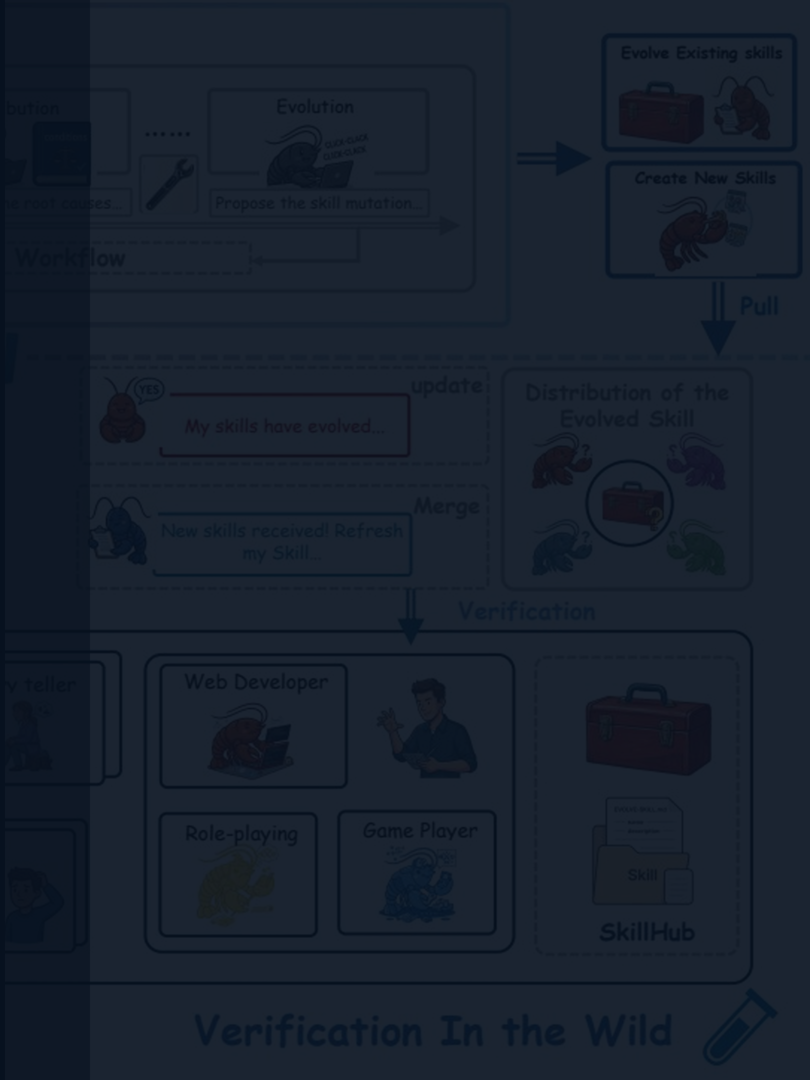
SKILLCLAW

Let Skills Evolve Collectively with Agentic Evolver

Collective skill evolution for multi-user LLM agent ecosystems

Interaction trajectories become shared evidence for verified skill updates.

arxiv.org/abs/2604.08377



PROBLEM FRAMING

Agent Skills Remain Static After Deployment — Users Rediscover Solutions Independently

Static skill paradigm

- Manual install from hub
- Runtime solutions stay local
- Failures recur across users
- No system-level accumulation

Evolving skill paradigm

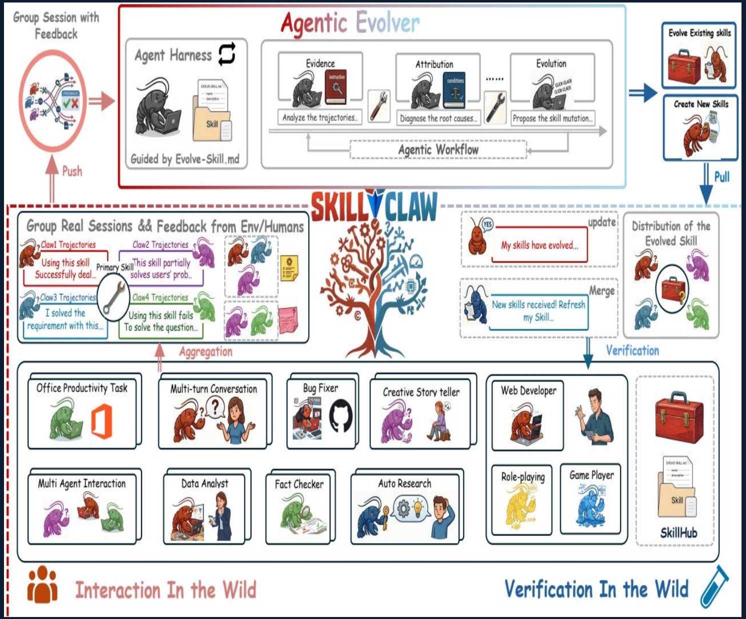
- Collect trajectories
- Generalize beyond one instance
- Update structured skills
- Synchronize improvements

Memory stores trajectories but remains instance-tied; skill libraries compress experience but are static.

Source: arxiv.org/abs/2604.08377

ARCHITECTURE OVERVIEW

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution



Overview figure

01
Evidence
Group skill-referenced trajectories

02
Attribution
Diagnose failures and successes

03
Evolution
Propose verified skill mutations

04
Distribution
Synchronize shared repository

Multi-user Interaction → Session Collection → Skill Evolution → Skill Synchronization

Updated skills are verified before merge and synchronized back to all agents.

Source: arxiv.org/abs/2604.08377

Agentic Evolver: Evidence → Attribution → Evolution in Three Stages

1 Evidence

Analyze trajectories to identify behavioral patterns across action–feedback causal chains.

2 Attribution

Diagnose root causes behind failures and successes, separating symptoms from causes.

3 Evolution

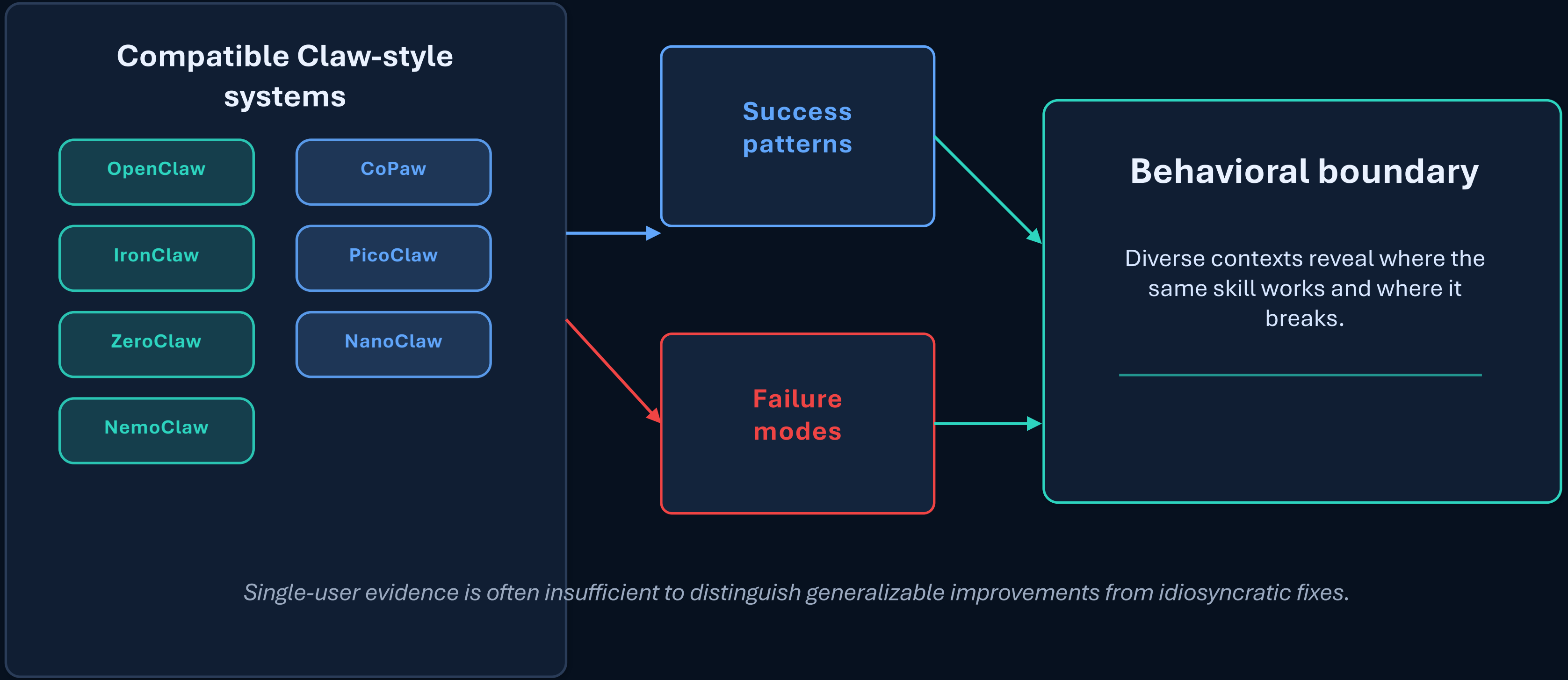
Propose skill mutations that refine existing skills or create new ones.

The evolver is an autonomous agent performing open-ended reasoning over interaction evidence — not a predefined rule system.

Source: arxiv.org/abs/2604.08377

COLLECTIVE EVIDENCE

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries



Single-user evidence is often insufficient to distinguish generalizable improvements from idiosyncratic fixes.

Source: arxiv.org/abs/2604.08377

CASE STUDY 1

Case Study: Multimodal Creative Pipeline Skill — 6 Days of Evolution Attempts

Validator behavior: conservative admission, with only the Day 1 candidate entering the best pool.

1 of 6 accepted

Date	Outcome	Brief rationale
Day 1	Accepted	validate-tmp-workspace-inputs: check /tmp_workspace inputs & environment setup; upgrade to new best pool
Day 2	Rejected	multimodal-input-validation-and-setup: validation & output env init; keep current best pool
Day 3	Rejected	multimodal-creative-task-pipeline: extract from PDF/video/image; keep current best pool
Day 4	Rejected	pipeline improvement: image classification, visual generation, structured validation; keep current best pool
Day 5	Rejected	pipeline + validate-required-input-files: per-file fail-fast validation; keep current best pool
Day 6	Rejected	candidate extended PDF-to-poster / document-to-visual paths; not admitted to next cycle

Source: arxiv.org/abs/2604.08377

CASE STUDY 2

Case Study: Git Push Auth-Fallback Skill — Iterative Refinement Over 6 Days

Contrast: the skill improved across three accepted update days before reaching a plateau.

3 of 6 accepted updates

Date	Outcome	Brief rationale
Day 1	Accepted	safe fallback instead of blocking on push failure; add to Safety best pool
Day 2	Accepted	unified patch/bundle filenames, reduced inconsistency; keep updated best pool
Day 3	Accepted	auth-alternative paths & secrets audit; fixed subshell pitfalls; keep current best pool
Day 4	No update	same-pool retest; validator confirmed current pool as best; continue same best pool
Day 5	Rejected	push hang treated as auth failure; no improvement; keep current best pool
Day 6	Rejected	identity config & filename consistency; no improvement; not admitted to next cycle

Day 4 was an accepted same-pool retest, not one of the three accepted updates.

Source: arxiv.org/abs/2604.08377

Key Takeaways: Three Properties That Distinguish SkillClaw from Existing Systems

01

Collective evolution

Individual interactions improve a shared, continuously improving skill ecosystem.

02

Fully automatic

Evolution is driven by runtime interaction without manual curation or explicit user intervention.

03

Agentic evolution paradigm

Updates are produced through open-ended reasoning, not predefined update rules.

Implication: agent systems can move from static skill hubs toward continuously improving ecosystems.

Source: arxiv.org/abs/2604.08377